

HCI & Computer Graphics

HCI Focus: Interaction Elements :

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# Figure 1 — HCI interaction elementsimport matplotlib
.pyplot as plt
import matplotlib.patches as patches
# Use your Google Earth screenshot
image = plt.imread("google_earth_screenshot.png")
fig, ax = plt.subplots(figsize=(12, 7))ax.imshow(image)
ax.axis("off")
# 1. Search box
ax.add_patch(
    patches.Rectangle(
        (0.03, 0.05), 0.28, 0.08,
        transform=ax.transAxes,
        fill=False, linewidth=2
    )
)
ax.annotate(
    "1. Search box",
    xy=(0.17, 0.09), xycoords="axes fraction",
    xytext=(0.38, 0.03), textcoords="axes fraction", arrowprops=dict
    (arrowstyle="->", linewidth=2),
    fontsize=12, fontweight="bold"
)
# 2. Zoom control
ax.add_patch(
    patches.Rectangle(
        (0.91, 0.36), 0.06, 0.18,
        transform=ax.transAxes,
        fill=False, linewidth=2
    )
)
ax.annotate(
    "2. Zoom control",
    xy=(0.94, 0.45), xycoords="axes fraction",
    xytext=(0.70, 0.30), textcoords="axes fraction", arrowprops=dict
    (arrowstyle="->", linewidth=2),
    fontsize=12, fontweight="bold"
)
# 3. Navigation / rotate control
ax.add_patch(
    patches.Circle(
        (0.94, 0.73), 0.07,
        transform=ax.transAxes,
        fill=False, linewidth=2
    )
)
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)
)
ax.annotate(
    "3. Navigation / rotate control",
    xy=(0.94, 0.73), xycoords="axes fraction",
    xytext=(0.58, 0.82), textcoords="axes fraction", arrowprops=dict
    (arrowstyle="->", linewidth=2),
    fontsize=12, fontweight="bold"
)plt.tight_layout()
plt.savefig("figure1_hci_elements.png", dpi=300, bbox_inches="tight")plt.show()

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Figure 2 — Computer Graphics Focus: Visual / Rendering Elements

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# Figure 2 — Computer Graphics visual / rendering elementsimport matplotlib.pyplot
as plt
import matplotlib.patches as patches
# Use your Google Earth screenshot
image = plt.imread("google_earth_screenshot.png")
fig, ax = plt.subplots(figsize=(12, 7))ax.imshow(image)
ax.axis("off")
# 1. 3D terrain / mesh
ax.add_patch(
    patches.Rectangle(
        (0.28, 0.40), 0.38, 0.30,
        transform=ax.transAxes,
        fill=False, linewidth=2
    )
)
ax.annotate(
    "1. 3D terrain / mesh",
    xy=(0.47, 0.55), xycoords="axes fraction",
    xytext=(0.05, 0.72), textcoords="axes fraction", arrowprops=dict
    (arrowstyle="->", linewidth=2),
    fontsize=12, fontweight="bold"
)
# 2. Lighting & shadows
ax.add_patch(
    patches.Ellipse(
        (0.72, 0.57), 0.18, 0.12,
        transform=ax.transAxes,
        fill=False, linewidth=2
    )
)
ax.annotate(
    "2. Lighting & shadows",
    xy=(0.72, 0.57), xycoords="axes fraction",

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xytext=(0.73, 0.76), textcoords="axes fraction", arrowprops=dict
(arrowstyle="->", linewidth=2),
fontsize=12, fontweight="bold"
)
# 3. Perspective / depth
ax.add_patch(
patches.Rectangle(
(0.48, 0.18), 0.20, 0.17,
transform=ax.transAxes,
fill=False, linewidth=2
)
)
ax.annotate(
"3. Perspective / depth",
xy=(0.58, 0.26), xycoords="axes fraction",
xytext=(0.67, 0.12), textcoords="axes fraction", arrowprops=dict
(arrowstyle="->", linewidth=2),
fontsize=12, fontweight="bold"
)
plt.tight_layout()
plt.savefig("figure2_graphics_elements.png", dpi=300,
bbox_inches="tight")plt.show()

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